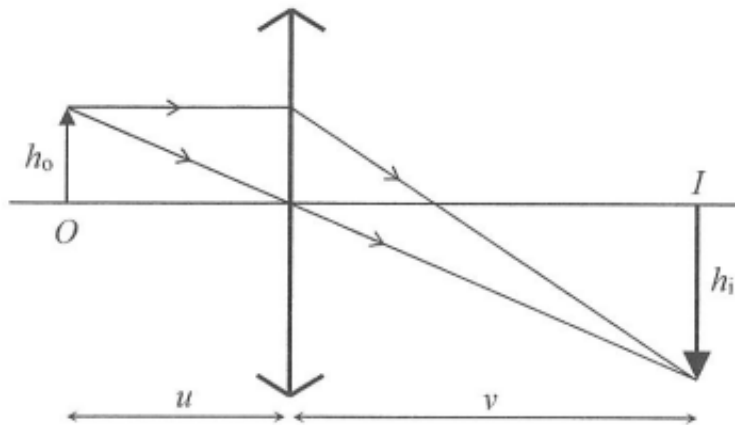


Magnification



Linear magnification (M)

$$M = \frac{v}{u} = \frac{h_i}{h_o}$$

v : image distance

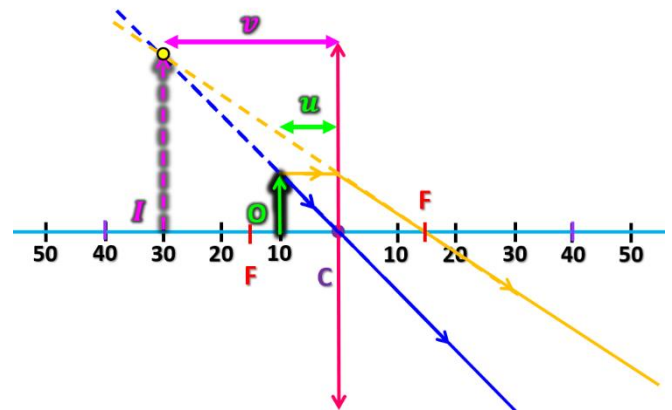
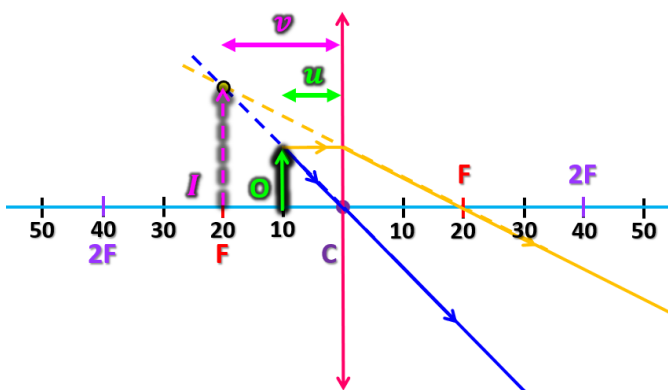
u : object distance

h_i : image height

h_o : object height

$$\text{Magnification} = \frac{I}{O} = 2 = \frac{v}{u}$$

$$\text{Magnification} = \frac{I}{O} = 3 = \frac{v}{u}$$



Lens formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

v : image distance (**real image: +ve, virtual image: -ve**)

u : object distance (**always +ve**)

f : focal length (**convex lens: +ve, concave lens: -ve**)

1. An object placed in front of a converging lens of focal length 10 cm forms an image 30 cm from the lens. If the image is real, what is the distance of the object from the lens?

- A. 7.5 cm
B. 8.5 cm
C. 12.5 cm
D. 15 cm

D
$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$
$$\Rightarrow \frac{1}{u} + \frac{1}{+30} = \frac{1}{+10}$$
$$\Rightarrow u = +15 \text{ cm}$$
 (Image distance is positive as the image is real.)

D

2. An object placed in front of a converging lens of focal length 10 cm forms an image 30 cm from the lens. If the image is virtual, what is the distance of the object from the lens?

- A. 7.5 cm
B. 8.5 cm
C. 12.5 cm
D. 15 cm

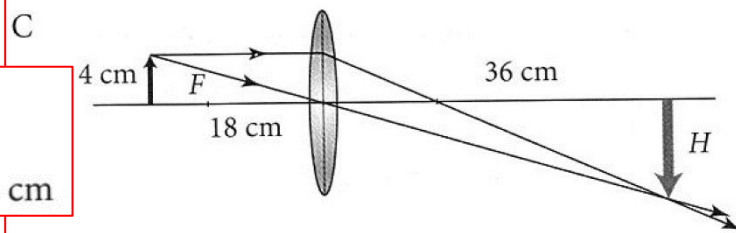
A
$$\frac{1}{u} + \frac{1}{-30} = \frac{1}{+10}$$
$$\Rightarrow u = +7.5 \text{ cm}$$
 (Image distance is negative as the image is virtual.)

A

3. An object of height 4 cm is placed 18 cm in front of convex lens of focal length 12 cm. A real image is formed 36 cm on the other side of the lens. What is the height of the image?

- A. 2 cm
B. 4 cm
C. 8 cm
D. 12 cm

$$\frac{H}{4} = \frac{36}{18}$$
$$\Rightarrow H = 8 \text{ cm}$$



C

4. An illuminated object placed 10 cm in front of a convex lens of focal length 8 cm forms a 4 times linearly magnified image on a screen. What is the distance between the object and the image?

- A. 32 cm
B. 40 cm
C. 48 cm
D. 50 cm

D Linear magnification = 4
$$\Rightarrow \frac{v}{u} = 4$$
$$\Rightarrow v = 4u = 4 \times 10 = 40 \text{ cm}$$
Distance between object and screen = $u + v = 50 \text{ cm}$

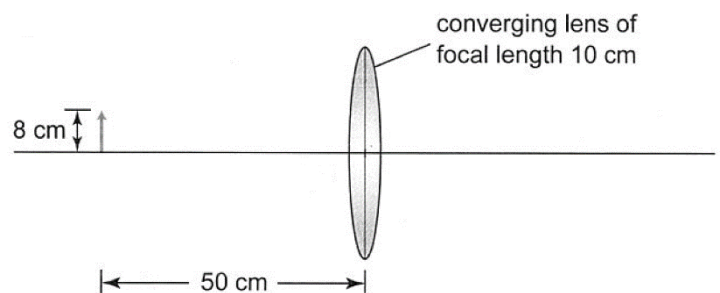
D

5. An object of height 8 cm is placed 50 cm from a converging lens of focal length 10 cm. What is the height of the image of the object?

- A. 2 cm
B. 3 cm
C. 4 cm
D. 5 cm

A
$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$
$$\Rightarrow \frac{1}{+50} + \frac{1}{v} = \frac{1}{+10}$$
$$\Rightarrow v = +12.5 \text{ cm}$$

$$m = \left| \frac{v}{u} \right| = \frac{12.5}{50} = 0.25$$
 (m is the linear magnification of the image.)
$$\Rightarrow \frac{H}{h} = 0.25$$
 (H and h are the heights of the image and the object respectively.)
$$\Rightarrow H = 0.25 \times 8 = 2 \text{ cm}$$



A

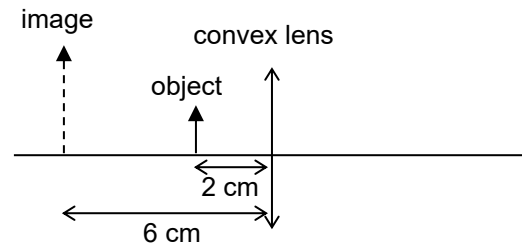
6. In the figure below, the positions of an object and its image formed by a convex lens are shown. The focal length of the lens is

- A. $\frac{1}{2} + \frac{1}{-6}$ cm
B. $\frac{(-6) \times 2}{-6+2}$ cm
C. $\frac{6 \times 2}{6+2}$ cm
D. $\frac{1}{2} + \frac{1}{6}$ cm

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{f} = \frac{1}{2} - \frac{1}{6}$$

$$f = 3 \text{ cm}$$



B

7. A student holds a magnifying glass 3 cm above a piece of paper on which the letters 'AT' is shown in the figure below. The magnification is 3. Find the image distance and the focal length of the lens.

Image distance **Focal length of the lens**

- A. 1 cm 0.75 cm
B. 1 cm 1.5 cm
C. 9 cm 2.25 cm
D. 9 cm 4.5 cm

$$m = \frac{v}{u}$$

$$3 = \frac{v}{3}$$

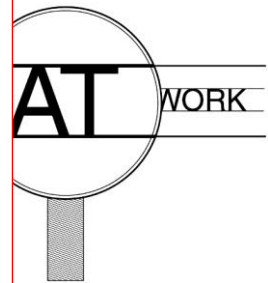
$$v = 9 \text{ cm}$$

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{3} - \frac{1}{9} = \frac{1}{f}$$

$$f = 4.5 \text{ cm}$$

the letters 'AT' is



D

8. The magnification of an image is 2, and the distance between the image and the lens is 8 cm. Which of the following may be the distance between the image and the object?

- (1) 4 cm
(2) 8 cm
(3) 12 cm
A (3) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

By $m = \frac{v}{u}$,
 $u = \frac{v}{m} = \frac{8}{2} = 4 \text{ cm}$
Since the image is magnified, the lens must be convex. There are two scenarios the image can be formed.

Scenario 1:
If the object distance is between f and $2f$, the image and the object are on opposite sides of the lens.
Distance between the image and the object = $4 + 8 = 12 \text{ cm}$

Scenario 2:
If the object distance is smaller than f , the image and the object are on the same side of the lens.
Distance between the image and the object = $8 - 4 = 4 \text{ cm}$
 $\therefore$ (1) and (3) are correct. (2) is incorrect.

B

9. An object is placed 20 cm in front of a converging lens of focal length 30 cm. Which of the following statements about its images is/are correct (1):

- (1) The image is real.
(2) The image is magnified.
(3) The image is erect.
A. (1) only
B. (1) and (2)
C. (2) and (3)
D. (1), (2) and (3)

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{20} + \frac{1}{v} = \frac{1}{30}$$

$$v = -60$$

Thus, the image is virtual.

(2): The magnification of the object is

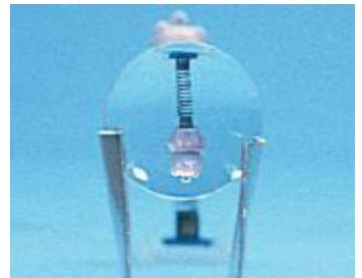
$$= \frac{60}{20} = 3$$

(3): Since the object distance is less than 1 focal length, the image is a magnifying and erect one.

C

10. A toy is viewed through a convex lens as shown.

- State the nature of the image.
- If the object and image distances are 10 cm and 5 cm respectively, find the magnification of the image.



(a) Real, inverted, diminished

$$\begin{aligned} \text{(b) Magnification} &= \frac{v}{u} \\ &= \frac{5}{10} \\ &= 0.5 \end{aligned}$$

11. In the figure below, a woman holds a lens in front of her right eye.

- What type of lens does the woman hold? Explain your answer.
- Suppose the distance between the lens and the woman's right eye is 6 cm and the focal length of the lens is 12 cm. Find the image distance.

(a) Since the image is magnified, the lens is a convex lens.

(b) Applying the lens formula,

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{6} + \frac{1}{v} = \frac{1}{12}$$

$$v = -12$$

$\therefore$ The image distance is 12 cm.



12. A folding magnifier with a magnification of 4 is shown below. It consists of a single lens supported by a stand of height 9 cm. The object to be viewed is placed below a square opening.

- State the nature of the image formed (erect or inverted, real or virtual).
- Find the image distance and the focal length of the lens.

(a) The image is erect and virtual.

(b) By $m = \frac{v}{u}$,

$$\text{image distance } v = mu = 4 \times 9 = 36 \text{ cm}$$

Apply the lens formula,

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{9} + \frac{1}{-36} = \frac{1}{f}$$

$$f = 12$$

$\therefore$ The focal length of the lens is 12 cm.



販賣,違者必究

13. An object is placed 12 cm from a convex lens. A clear image of the object is form on a screen 5 cm away from the lens.

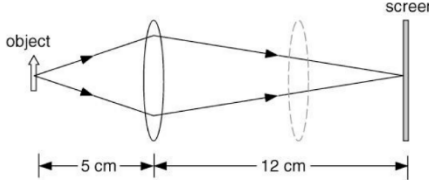
(a) Find the magnification of the image. (2 marks)

(b) (i) Suppose the object and the screen are fixed while the lens is moved towards the object. How far should the lens be moved so that a sharp image can be caught on the screen again? Explain your answer briefly. (3 marks)

(ii) Find the magnification of this new image. (1 mark)

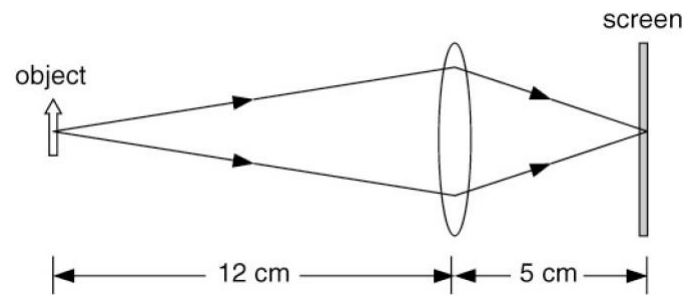
(a) Magnification = $\frac{\text{image distance}}{\text{object distance}}$
 $= \frac{5}{12}$
 $= 0.417$

(b) (i) If we view the screen through the lens, its image will be formed at the position of the object.
 Therefore, when the object distance is 5 cm, the image distance will be 12 cm.



Distance moved by the lens = $12 - 5 = 7$ cm

(ii) Magnification = $\frac{\text{image distance}}{\text{object distance}} = \frac{12}{5} = 2.4$



14. In the figure below, an illuminated letter 'L' is placed 30 cm in front of a convex lens and its image formed by the lens is caught by a translucent screen. The image is half as large as the object.

(a) (i) Is the image real or virtual? Explain your answer.

(ii) Sketch the shape of the image seen by the observer.

(b) Find the image distance and the focal length of the lens.

(a) (i) The image is real since it can be captured by a screen.

(ii) (Correct image)



(b) By $m = \frac{v}{u}$,

$$\text{image distance } v = mu = \frac{1}{2} \times 30 = 15 \text{ cm}$$

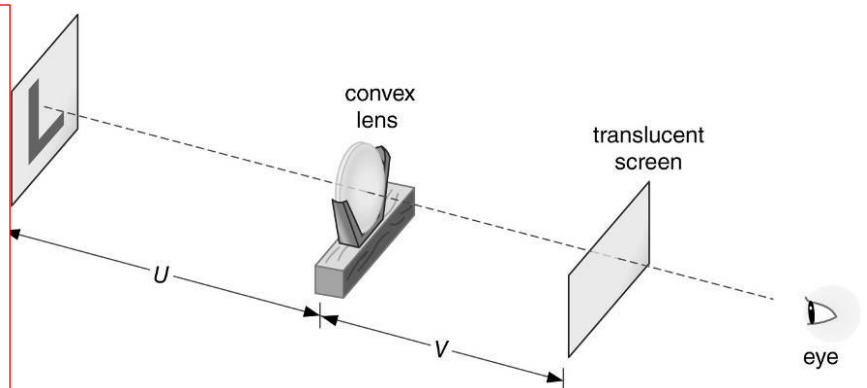
Applying the lens formula,

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{30} + \frac{1}{15} = \frac{1}{f}$$

$$f = 10$$

$\therefore$ The focal length of the lens is 10 cm.



15. Anthony uses the set-up shown in Figure a to study the image formation of a convex lens. He puts an illuminated letter F in front of the lens and an image is formed on a translucent screen. He finds that the object distance u is larger than the image distance v .

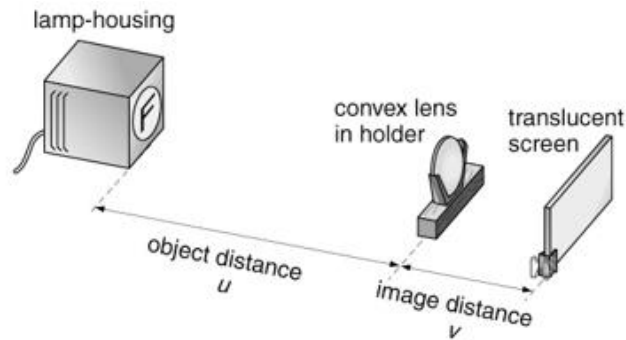


Fig a

- Sketch the image on the screen as observed from the right side of the set-up.
- Is the image real or virtual? Explain your answer.
- Anthony changes the object distance u and measures the values of v . After obtaining several sets of data, he constructs the graph shown in Figure b.
 - Find the magnification of the image when $u = 20$ cm.
 - If the object is moved closer to the lens, how will the image size change?
 - Find the value of u when $u = v$. Hence find the focal length of the lens.
- What will happen to the image if the upper half of the lens is covered by a piece of opaque paper?
- Explain, with the help of a ray diagram, why an image cannot be formed on the screen when the object distance is shorter than the focal length of the lens.

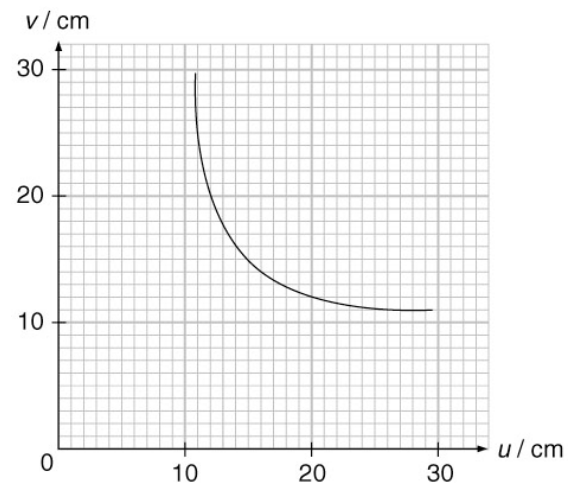


Fig b

(Dotted lines to show the virtual property)↗

16. An illuminated object is placed in front of a lens and a sharp image is formed on a screen on the other side of the lens. The magnification of the image is 4. The focal length of the lens is 20 cm.

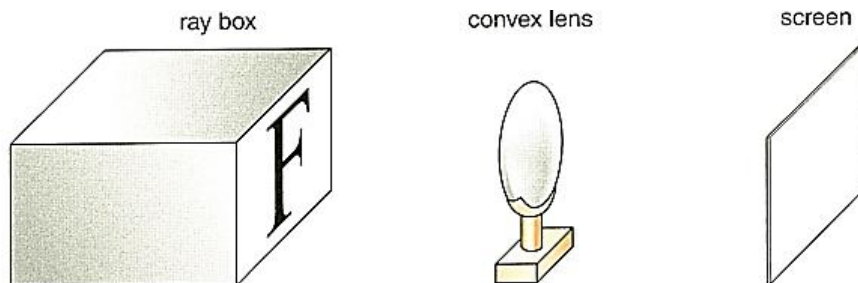
- (a) What type of lens is used? Explain briefly.
(b) Find the object and the image distances.

(a) The lens is a convex lens
since the image is real.

(b) By $m = \frac{v}{u}$,
 $v = mu = 4u$
Put $v = 4u$ and $f = 20$ into the lens formula,
$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$
$$\frac{1}{u} + \frac{1}{4u} = \frac{1}{20}$$
$$u = 25$$

$\therefore$ The object distance is 25 cm.
Image distance = $mu = 4 \times 25 = 100$ cm

17. The figure below shows the set-up of an experiment. An image of a letter 'F' is formed on a screen by a convex lens.



- (a) Is the image real or virtual? (1 mark)
(b) If the convex lens is equidistant from the ray box and the screen, what is the size of the image relative to the object? (1 mark)
(c) If the set-up is placed underwater, how will the image distance change? Explain briefly. (4 marks)

(a) Real
(b) Same size as object
(c) The image distance increases.
This is because the refractive index of water is higher than that of air.
Light is refracted by a smaller amount when it passes the water-lens interface and so the focal length of the lens increases.